MIL-H-40082A
30 September 1968
SUPERSEDING
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MILITARY SPECIFICATION

HUBS AND DIES, FOR HERALDIC ITEMS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers steel hubs and dies primarily for extracting working dies used in the manufacture of heraldic items (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of standards required by contractors in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design.- The heraldic design shall be as shown on the furnished applicable drawing, plaster model, galvano or other hub or die as specified (see 6.2).

3.2 Lead strikes.- Lead strikes from the die shall be submitted for Government approval and acceptance before hubs and dies are hardened.

FSC 8455

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3.3 Material.-

3.3.1 Steel.- The steel for the hubs and dies shall be either a high carbon tool steel, a special purpose steel or a shock resistant steel suitable for the purpose. Steel designated by the American Iron and Steel Institute (AISI) as carbon tool steels W1, W2 or W4 and of one of the following grades 1, 2, best, special or extra are acceptable. Other special formulations some of which have not been yet classified by AISI but are commonly used within the industry for this purpose are acceptable. The following is a list of steels known to be suitable listed by their trade name or AISI designation.

	C	Mn	Si	Cr	V	Mo	W	Remainder
BFW	.45	.35	1.50	1.60	.25	-	-	Fe
TUSCA	.50	.60	.30	1.30	.25	.30	-	Fe
AISI Type S2	.50	.40	1.00	-	-	.50	-	Fe
AISI Type O1 (Ketos)	.90	1.35	.35	.50	-	-	.50	Fe
MYA	.42	.30	1.45	1.45	.25	-	-	Fe

These steels may be accepted on the manufacturer's certification.

3.4 Construction.- Hubs and dies shall be fabricated from steel specified in 3.3. Unless otherwise specified (see 6.2), the shape of the hubs and dies shall be round, square or oblong. The heraldic design shall be an exact replica of the design furnished by the Government, except, when required, of a specified reduced size (see 6.2).

3.4.1 Hub and die sizes.- Minimum height of hubs and dies shall be 2 inches except for heraldic designs 4 inches and over in length, width or diameter, in which case the minimun height shall be 2-1/2 inches. Minimum length, width or diameter shall be such as to provide a minimum of 3/8 inch clearance between edge and design for hubs and 1/2 inch minimum clearance between edge and design for dies. There shall be sufficient cut-away or relieved area around the embossing on each hub to insure penetration past the embossed design when driving the hub into the working die.

3.4.2 Hardening.- Hubs and dies shall be normalized, hardened and tempered. Hardness shall be Rockwell 58 to 63 C scale when tested as specified in 4.4.

3.5 Finish.- The top and bottom surfaces of all dies shall be ground parallel. The bottom surface of all hubs shall be ground smooth. The top surface shall have a minimum of three areas spaced 120 degrees apart, ground parallel to the bottom surface for hardness testing.

3.6 Marking for identification.- Hubs and dies shall be legibly marked on the outer perifhery by engraving or metal stamping with the following information.

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(a) Type of steel.
(b) Date hubs or dies were completed.
(c) Name or trade mark of manufacturer.
(d) Letters U.S. in characters not less than 1/8 inch high.

3.7 Workmanship.- The hubs and dies shall be clean, well made and free from imperfections which may adversely affect their durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Pre-hardening inspection.- Inspection of the lead strikes specified in 3.2 shall be accomplished to determine conformance with the heraldic design requirements specified in 3.4.

4.3 Inspection for acceptance.-

4.3.1 Certification of compliance.- Components and materials listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in applicable paragraphs of this specification.

Component	Characteristics	Rqm't para.
Steel	Material identification	3.3

4.3.2 Inspection of the end item.-

4.3.2.1 Visual examination of hubs and dies.- The defects found during examination shall be classified in accordance with 4.3.2.1.1 and 4.3.2.1.2. The hubs and dies shall be inspected in accordance with 4.3.2.1.3. The unit of product for these examinations shall be one hub or die.

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4.3.2.1.1 Examination of hubs or dies for defects in design, material, construction and workmanship.-

Examine	Defect
Design	Any characteristic not conforming to applicable drawing, plaster model, galvano or other hub or die as specified.
	Any significant detail not clear, marred, missing, reduced or obliterated; i.e., does not compare favorably with applicable drawing, plaster model, galvano or other hub or die as specified.
	Heraldic design not an exact replica of the Government furnished design or when required, not of a specified reduced size.
Construction	Not shaped as specified; i.e., not round, square, oblong or as otherwise specified.
	Driving room on hub omitted.
Workmanship	Mutilated or not free from imperfections such as scratches, tool marks, dig, gouge, dent, burr, splinter, rough edges, sharp fin or sharp edge which may effect durability or serviceability.
	Fracture, rupture, crack or broken.
	Malformed, damaged, warped, twisted, distorted, bent out of shape or otherwise impaired to an extent affecting durability or serviceability.
Quality of metal	Not fabricated from specified material.
	Surface spotted or open grained, i.e., scale, corrosion, rust or sludge deposits.
Cleanness	Not clean; i.e., scale, corrosion, rust or sludge deposits.
Identification marking	Missing, incorrect, illegible or text incomplete.
	Not marked by engraving or metal stamping.
	Letters U.S. in characters less than 1/8 inch high.
Finish	Surfaces not ground as required.

4.3.2.1.2 Examination of hubs and dies for defects in dimentions.- Any dimensions that is not within specified tolerances shall be classified as a defect.

4.3.2.1.3 Inspection level.- One hundred percent inspection shall be made for defects applicable to 4.3.2.1.1, 4.3.2.1.2 and 4.3.2.2. The hub or die shall be rejected if any defect is found.

4.3.2.2 Testing of the end item.- Testing of the completely fabricated hub or die shall be performed in accordance with table I for the characteristics shown therein. The sample unit shall be one completely fabricated hub or die.

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TABLE I End item testing.-

Characteristic	Rqm't para	Test method	Rqm't applicable to s/u	#determ per s/u	Results reported as
Hardness 1/	3.4.2	4.4	x	2	To nearest whole number

1/ Care shall be exercised to limit hardness readings to areas wherein indentations will not be harmful to the heraldic design.

4.4 Tests.-

4.4.1 Hardness test.- The hardness specified in 3.4.2 shall be determined by the Rockwell hardness test, method 243 of Fed. Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level C.-

5.1.1 Level C.- Hubs and dies shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving station. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level C.

5.2.1 Level C.- Hubs and dies packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The hubs and dies covered by this specification are intended to be loaned to contractors for extracting their working dies in the manufacture of heraldic items. This specification also covers master dies to be retained by the Government and used for replacement of Government hubs.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether drawing, plaster model, galvano or hub or die is to be furnished (see 3.1).
(c) Shape of hub or die, if other than specified (see 3.4).
(d) Size of heraldic design when reduction is required (see (3.4).

Custodians:
Army - IH
Air Force - 11
Navy - SA
Review Activities:
Air Force - 11,82,85
Navy - SA,MC
Other - DP
User Activity:
Army - SM

Preparing Activity:

The Institute of Heraldry,
United States Army

Project No. 8455-0128

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